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Author manuscript
Psychooncology. Author manuscript; available in PMC 2023 April 01.

Published in final edited form as:
Psychooncology. 2022 April ; 31(4): 597–605. doi:10.1002/pon.5843.

Stakeholder-informed conceptual framework for financial burden among adolescents and young adults with cancer

Suzanne C. Danhauer[1], **Mollie Canzona**[2], **Reginald D. Tucker-Seeley**[3], **Bryce B. Reeve**[4], **Chandylen L. Nightingale**[1], **Dianna S. Howard**[5], **Nicole Puccinelli-Ortega**[1], **Denisha Little-Greene**[1], **John M. Salsman**[1]

[1]Department of Social Sciences & Health Policy, Wake Forest School of Medicine, Winston Salem, North Carolina, USA

[2]Department of Communication, Wake Forest University, Winston Salem, North Carolina, USA

[3]Leonard Davis School of Gerontology, University of Southern California, Los Angeles, California, USA

[4]Department of Population Health Sciences, Duke University School of Medicine, Durham, North Carolina, USA

[5]Department of Internal Medicine, Section on Hematology and Oncology, Wake Forest School of Medicine, Winston Salem, North Carolina, USA

Abstract

Background: Cancer and its treatments can result in substantial financial burden that may be especially distressing for adolescents and young adults (AYAs) since they are at a developmental stage focused on completing one's education and establishing independence. The purpose of this study was to develop a conceptual model of financial burden among AYA cancer patients to inform development of a financial burden measure.

Methods: In-depth concept elicitation interviews were conducted with a purposive-selected stakeholder sample (36 AYAs and 36 AYA oncology health care providers). The constant comparative method was used to identify themes that illustrate AYAs' experience of financial burden by stakeholder groups.

Results: Eleven financial burden themes emerged: (1) impact of socioeconomic status and age; (2) significant cancer costs; (3) indirect cost "ripple effects"; (4) limited awareness of costs (adolescents); (5) emotional impact; (6) feeling over-whelmed navigating the health care

Correspondence: Suzanne C. Danhauer, Department of Social Sciences & Health Policy, Wake Forest School of Medicine, Winston Salem, NC 27157, USA. danhauer@wakehealth.edu.

AUTHOR CONTRIBUTIONS
Conceptualization, design: Salsman, Danhauer, Canzona, Reeve. Methodology: Salsman, Canzona, Danhauer, Reeve, Tucker-Seeley, Little-Green. Analysis: Canzona, Puccinelli-Ortega. Data interpretation and results: Danhauer, Tucker-Seeley, Canzona, Reeve, Salsman, Nightingale, Howard. Writing, editing, approval of final manuscript: All authors.

CONFLICTS OF INTEREST
The authors have no conflicts of interest, financial or otherwise, related to the content of this manuscript.

SUPPORTING INFORMATION
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system; (7) treatment decision modifications; (8) reducing spending; (9) coping strategies; (10) financial support; and (11) long-lasting impact. The conceptual model highlights the importance of material, psychosocial, and behavioral domains of financial burden with an emphasis on phase along the cancer continuum and developmental stage in the experience of financial burden for AYAs.

Conclusions: Issues presented in the voice of AYA patients and providers highlight the profound impact of financial burden in this survivor group. The next step in this work will be to develop and test a patient-reported measure of financial burden among AYA cancer survivors.

Keywords

adolescent; adolescent and young adult oncology; cancer; financial burden; financial distress; financial toxicity; oncology; psycho-oncology; stakeholder input; young adult

1 | INTRODUCTION

There are nearly 2 million people in the United States who are living with or have survived cancer as an adolescent or young adult (AYA; ages 15–39 years).[1] Self-reported financial burden is disproportionately greater in AYAs with cancer than any other age group with cancer.[2–4] A cancer diagnosis and treatment for AYAs may be especially disruptive given unique emotional and social life changes that take place during adolescence and young adulthood (e.g., making decisions about education, choosing a career) that can impact financial burden.[5–7] Further, AYAs with cancer may have inadequate insurance coverage, limited financial assets, and significant work interruption, leading to greater financial burden during and after treatment.[8–11]

Despite this disproportionate impact of financial burden, most health-related quality of life (HRQOL) measures fail to capture the domains most relevant to AYA cancer survivors, including financial burden.[10,12] In a descriptive, pilot study of AYA cancer survivors to examine HRQOL across treatment type (“on” vs. “off”) and developmental age cohort (adolescents, emerging adults, young adults), our group found that financial issues were one of the most important HRQOL domains reported by AYA survivors.[13,14] Similarly, in a population-based observational study of AYAs conducted by the U.S. National Cancer Institute, financial concerns were one of the most frequently reported negative impacts of cancer regardless of developmental subgroup.[7]

Few research studies have explored the confluence of the unique developmental issues of adolescence/young adulthood and the financial burden of cancer survivorship that AYA cancer survivors must navigate/manage. Our team’s overarching goal is to give AYAs “a voice” in this important and challenging aspect of their experience with cancer and to ultimately improve and optimize patient-centered care. More specifically, we are leveraging the scientific standards of the Patient-Reported Outcomes Measurement Information System®[15] to develop a psychometrically robust measure of financial burden for AYAs with cancer and their caregivers. The first step of this process, as detailed in this manuscript, is a rigorous qualitative approach that involved developing and refining a

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conceptual frame-work for financial burden through in-depth, qualitative concept elicitation interviews with key stakeholders.

2 | METHODS

2.1 | Creation of preliminary conceptual framework prior to concept elicitation interviews

We first developed a preliminary conceptual framework for financial burden, guided by a literature review, our project team discussions, and interviews with content experts. Content experts included professionals with financial burden expertise such as clinicians, researchers, and those working in leadership roles within relevant non-profit, patient-support organizations (e.g., American Cancer Society, Samfund) in addition to medical schools, health systems, and major medical centers. This preliminary framework drew extensively from the research in cancer survivorship that suggests that financial hardship is experienced in material (e.g., out-of-pocket medical expenses, reduction in income/assets, medical debt, bankruptcy), psychosocial (i.e., stress, distress, and worry about medical costs or concerns about work impairment associated with cancer), and behavioral domains (coping behaviors and strategies such as delaying or forgoing medical care).[16–19] While the material, psychosocial, and behavioral domain model helps in sorting the many terms used to describe the financial hardship experience during survivorship, it was unclear if this model was sufficient to capture the unique experiences of AYAs as they navigate cancer survivorship. This preliminary framework informed the development of an in-depth interview script used in this study to elicit data to refine this framework.

2.2 | Eligibility & recruitment

Key stakeholders included AYA cancer survivors and a multidisciplinary group of health professionals who provide treatment or supportive services to this survivor group.

Adolescents or young adults were eligible if they: (1) were 15–39 years of age at diagnosis and currently; (2) were English-speaking; (3) had a cancer diagnosis, any type; and (4) were receiving active treatment or were 0–5 years post-treatment. The electronic medical record at each participating site was screened to identify potentially eligible patients. Permission to contact eligible AYAs (and parents of 15–17-year-olds) for participation was secured from treating physicians. Participants were recruited from the Wake Forest Baptist Comprehensive Cancer Center (WFBCCC) and the Robert H. Lurie Comprehensive Cancer Center of Northwestern University (RHLCCC). The study received institutional review board (IRB) approval at both of these sites (IRB00044525 at WFBCCC; STU00208207 at RHLCCC). All interested and eligible AYA participants provided written informed consent and were compensated $50 after finishing their interviews.

For AYA cancer providers, we were interested in a multidisciplinary group of health professionals, including those who provide supportive care to help address various needs of AYAs, including financial concerns. Additional inclusion criteria were: (1) provision of care to AYAs; (2) ≥2 years practicing; and (3) English-speaking. We recruited AYA providers by leveraging existing professional networks. More specifically, AYA providers were identified through national AYA initiatives or working groups (e.g., National Clinical Trials Network,

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Teen Cancer America). The study coordinator confirmed eligibility via e-mail or phone and obtained informed consent.

2.3 | Data collection

We have followed the Standards for Reporting Qualitative Research for study design, execution, and reporting.[20] All interviews were conducted in two waves through the Qualitative and Patient-Reported Outcomes (Q-PRO) Shared Resource in the WFBCCC by trained research associates with experience in qualitative methods. Patient and provider interview guides (see Supporting Information) included questions on financial burden. All interviews were audio recorded with permission.

2.3.1 | Patient interviews—In-depth interviews were conducted with 36 AYA patients (10 adolescent [15–17 years old]; 12 emerging adult [18–25 years old], and 14 young adult [26–39 years old]). Interviews with WFBCCC patients were conducted in-person ($n = 29$). Interviews with RHLCCC patients were conducted by video call ($n = 4$) or phone ($n = 3$). Interviews lasted an average of 60 min and were conducted between 13 February 2018 and 5 September 2019.

2.3.2 | Provider interviews—Semi-structured interviews were conducted by phone with 36 AYA providers to allow for a wide range of disciplines and mirror the multidisciplinary care AYAs receive (details in Table 1). Interviews lasted an average of 36 min and were conducted between 25 April 2018 and 28 June 2019.

2.3.3 | Qualitative analysis—All interviews were professionally transcribed, de-identified and reviewed against original audio to ensure accuracy. Transcripts were imported into Atlas.ti (Version 7.5) to store and manage the data. Interview responses were examined systematically.[21–23] Thematic analysis via the constant comparative method was used to analyze transcripts stratified by developmental stage (adolescents, emerging adults, young adults) and group (patient and provider). First, a code-book was developed inductively in collaboration between a study team investigator (MC) and Q-PRO. To create this codebook a total of 55% of interviews were cross coded by two Q-PRO research associates who met regularly to confirm reliability and resolve all coding discrepancies. The remaining transcripts were then independently coded by two Q-PRO research associates using the standardized codebook. The codebook was revised iteratively based on emergent themes. Changes made to codes and code definitions were documented as they were made. Constant comparative methods were used to assess similarities and differences of each interview with the previously identified themes. Results of stratified analysis were then systematically compared to identify areas of overlap and variation, collapsing and renaming thematic categories where appropriate and creating new theme labels for the final conceptualization of results when warranted. Q-PRO research associates met regularly with the study team investigator (MC) to discuss emerging concepts and themes.[24]

Data saturation was evaluated at the theme level for AYA and provider interviews. AYA interview transcripts were divided into sets of four by age group (adolescent, emerging adult, and young adult) and order that they were completed. Provider interview transcripts were

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divided into sets of four based on chronological order. Themes were saturated when the research team members considered the descriptions of financial burden to be similar in detail by multiple participants within each four-interview set.[21,22]

3 | RESULTS

Stakeholder characteristics are provided in Table 1. Patient's socio-economic status was not assessed for the adolescent group, as all were primarily dependent on parent(s) or a grandparent (and had little to no knowledge of family socioeconomic status). All emerging adults were financially dependent (75% [9/12] on their parents and 25% [3/12] on a spouse or someone else). In contrast, 50% (7/14) of the young adults were financially independent, while 43% (6/14) were financially dependent on a partner/spouse as the primary income earner. Of those who provided income, 14.28% earned <$15,000, 7.14% earned $15,000–$29,000, 21.42% earned $30,000–$59,999, 21.42% earned $60,000–$100,000, and 28.57% earned >$100,000. The majority of young adults were privately insured (57%), while 29% reported being insured by Medicaid.

Table S1 shows the following 11 financial burden themes that emerged from the in-depth interviews with the four stakeholder groups (adolescent, emerging adult, young adult, provider): (1) Impact of pre-cancer socioeconomic status and age on financial burden; (2) Significant direct and indirect costs of cancer and its treatment; (3) "Ripple effects" of indirect costs related to cancer; (4) Limited awareness of direct medical costs of cancer by adolescents; (5) Emotional impact of financial burden; (6) Sense of overwhelm navigating the health care system; (7) Modification of cancer treatment-related behaviors to reduce financial burden; (8) Adjustment of spending and organizational strategies to manage finances; (9) Use of psychological/behavioral strategies to cope with distress related to financial burden; (10) Financial support/services from organizations; and (11) Long-lasting impact of financial burden. Properties and exemplar quotes for each theme are provided in Table S1.

3.1 | AYA age group comparisons between themes

Several patterns emerged related to the various age groups of stakeholders: (1) Socioeconomic status affected participants' experiences with financial burden across all AYA age groups. Specifically, all adolescents were dependent on a caregiver (usually parent[s]) for insurance coverage and financial support, or had government-sponsored health coverage (Medicaid). Emerging adults reported varying degrees of financial independence and insurance coverage (parent-, spouse- or government-sponsored health coverage; no health insurance); however, most often they were financially supported by their parents. Young adults had varying sources of insurance coverage. Some provided health insurance to their partners or were covered under their partners' plans; others had dependents who relied on them for support. Some had government-sponsored health coverage or no insurance. (2) Compared to the younger age groups, young adults expressed the highest level of awareness and concern about financial burden. (3) Providers observed that adolescent patients received more financial assistance than older AYA cohorts because people tend to be more willing to donate to children with cancer than adults with cancer. The adolescent data support

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this claim as nearly every patient reported receiving substantial financial assistance from programs or their social networks.

3.2 | Patient-provider comparisons between themes

Some differences also emerged related to responses from patients and providers: **Providers** described financial burden as an invasive, multifactorial, long-term problem that affects future financial health, future opportunities, and family members. Providers tended to express broad, global views of this domain, perhaps because their reflections comprised their experiences with many patients. This perspective differs from a patient focus on more immediate needs. In addition, some AYA **patients** discussed the implications of cancer-related financial burden on their social lives. Financial burden required them to limit spending on social activities, which could feel isolating. Few providers mentioned this connection between financial burden and social life, so it may represent a gap in provider understanding of the patient experience.

Table 2 describes how the financial burden themes that emerged from the in-depth interviews each fit with the material, psychosocial, and behavioral domains of the proposed conceptual model drawn from Tucker-Seeley et al.'s framework.[23] In our initial model, the broad categories of material conditions (both resources and deficits) (e.g., direct and indirect medical costs, productivity loss, medical debt/bankruptcy), psychosocial responses (e.g., distress related to financial hardship), and behavioral strategies (e.g., cost-cutting strategies, cutting back on health care use) would impact patient and family health and well-being outcomes. Using our interview data, this model was expanded to reflect the multiple layers of impact and influence on financial burden to include AYA patients/families, health care providers, the broader health care system, and the health policy environment (see Figure 1).

Health care providers/team members were able to offer a broader perspective of financial burden encompassing experiences with many AYA patients (in contrast to a typically single patient/family experience heard in our patient stakeholder interviews). One striking idea that emerged from provider interviews was the potential for making oncology care and treatment decisions influenced by patient financial burden concerns, an issue that warrants discussion with patients and families. This point raises the question of whether financial burden concerns influence care/treatment decisions, care/treatment decisions influence financial burden, or both. In addition, while some AYAs described adequate insurance coverage, concerning system-level factors were identified as other AYAs reported inadequate insurance coverage and feeling overwhelmed navigating the health care system. This overwhelmed feeling may highlight the need for health care systems to streamline processes or make them more transparent to decrease this feeling in AYA patients and families. Finally, with regard to the health policy environment, providers noted some AYAs delay health care when uninsured or underinsured and the need for legislation to insure adequate health care for all AYAs. It is critical to examine health coverage and how it may be impacted by a cancer diagnosis at a young age, especially when patients would typically be transitioning to their own health coverage policies. Advocacy efforts will continue to be important to ensure that young adults are not substantially hindered in their pursuit of financial independence by significant medical debt due to no or poor health coverage.

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Based on the themes identified in the stakeholder interviews, it was clear that age/developmental stage (e.g., varied emotional reactions by age such as guilt that parents have to pay for adolescent's medical expenses compared to profound stress/anxiety thinking about medical bills for young adult survivors) and time along the cancer continuum (e.g., lasting financial effects of cancer treatment into post-treatment survivorship such as long-term debt and difficulty paying for basic life expenses) were two key factors that impacted many aspects of financial burden at multiple levels.

4 | DISCUSSION

Adolescents or young adults are at a disproportionately greater risk of financial burden than older adults with cancer. To our knowledge, this is the first study that conducted in-depth interviews with a broad representation of AYA cancer survivors and a multidisciplinary group of oncology providers to better inform our understanding of AYAs' experiences of financial burden. All participants responded to open-ended questions about their understanding of financial burden related to cancer and how it has impacted their lives in multiple ways. To develop a measure of financial burden specific to AYA cancer survivors, the goal of this investigation was to determine how qualitative data confirmed and extended an existing, multidimensional model of financial burden.[23,24]

Examination of multiple AYA patient and provider stakeholder groups identified 11 key themes related to financial burden. A review of the 11 themes showed material, psychosocial, and/or behavioral aspects related to each theme that were used to refine a conceptual model of financial burden in AYA cancer patients. The following discussion highlights the critical importance of hearing some of the most significant ways in which financial burden related to cancer has a profound impact on the lives of AYA cancer survivors and their families.

Adolescent or young adult cancer survivors are at a different developmental stage compared to their adult cancer survivor counterparts; the influence of both developmental processes and financial burden were revealed in our results. For example, AYA cancer survivors are generally in the early phases of their career, and cancer survivorship potentially stalls their career progression and leaves them behind their peers in terms of professional advancement and earnings.[10,25] In addition, cancer can be a particularly socially isolating experience for AYAs during a time when developing and deepening healthy relationships with others is critical. The potential for financial burden to further disrupt social connection among AYAs can compromise overall well-being and adversely impact quality of life.[26]

Theories of human development suggest that all cancer patients experience a similar set of life disruptions,[6] however, how those disruptions are managed and influence subsequent outcomes depends on when in the life course they occur and the resources one has to manage them. The themes in our study highlight the important developmental issues that can impact the financial burden of AYA survivors and should be explicitly recognized in future AYA cancer research studies such as when in the life course cancer survivorship is being managed, circumstances prior to diagnosis, and the enduring influence or "ripple effect" that managing a cancer diagnosis can have on longer-term financial resources.[27] Our

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themes demonstrated that the effects of financial burden certainly impact patients and family members; at the same time, the effects reach out further and impact oncology providers/care teams, health care systems, and ultimately state and federal health policy environments.

These findings have a number of important strengths. First and foremost, we have identified financial burden concerns among both AYA cancer survivors and providers that highlight gaps in previous findings and models. With our findings, we provide an expanded model that can be used to guide the development of a financial burden measure specific to the AYA cancer population. Further, our findings suggest that providers may be unaware of some impacts/concerns of financial burden among AYAs. Further education around this issue for providers may improve provider-patient communication and delivery of patient-centered care for AYAs with cancer. Finally, we had a diverse patient sample with respect to demographic and clinical characteristics that reflects the heterogeneity within each patient group. Similarly, the provider sample includes people from various roles and specialties (Table 1) with experience with various cancer populations (i.e., children, adults, military families, survivors of all ages). Their experiences and familiarity with each age group and domain varied. With these diverse groups of respondents, we reached thematic saturation within our overall groups and subgroups.

4.1 | Clinical implications

While the material-psychological-behavioral conceptual model of financial hardship recently proffered in previous cancer survivorship research has been useful for sorting the many terms used to describe the financial hardship experience,[16,23] it is evident from our results that for the AYA population, we need to include an explicit recognition of both developmental/life course (i.e., adolescence, emerging adulthood, or young adulthood) and time-related factors (i.e., where the cancer survivor falls along the cancer survivorship continuum) as well as the potential long-term influence that the financial burden of cancer can have on AYA survivors. Our results also highlighted the importance of the influence of multiple levels[24] including the patient/family, provider/care team, health care system/practice setting, and the state/federal health policy environment that can all impact the level of financial burden experienced by AYA cancer survivors.

4.2 | Study limitations

There are a few limitations to this work. First, in a study of financial burden, differences in material resources and insurance coverage are relevant. Our assessment of participants' financial material resources was limited and we were not able to do an in-depth examination of differences in experiences for those with ample versus those with fewer resources. Second, all of our participants were English-speaking and the majority of AYA participants were non-Hispanic white and came from the WFBCCC. These gender, race, and geographic distributions may limit the generalizability of our findings. That said, the WFBCCC serves a catchment area that is 98% medically underserved, 59% rural, and we recruited 40%–50% racial or ethnic minorities within each AYA age subgroup which likely provided greater socioeconomic diversity than many AYA oncology samples. Third, for adolescent and some emerging adult participants, we cannot confirm whether low perceived financial burden is an accurate perception or related to lack of awareness or understanding since they don't

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yet have financial responsibility for themselves. Fourth, we have not included data detailing the perspective of caregivers which is a critical piece of understanding financial burden in AYA cancer survivors and how parents/care-givers may provide a buffer against the impact of financial burden for younger AYA survivors.[28] Obtaining the caregiver perspective would provide a more complete perspective in these scenarios. Finally, there were no financial navigators in the provider group to provide any perspectives specific to dealing with AYA financial issues.

4.3 | Conclusions

Within the current dataset, we plan to examine themes around patient-provider communication as they were commonly mentioned in many of the interviews. We also will be able to explore dyadic perceptions with a subset of this sample for whom we have obtained data from eight AYA patient and family caregiver dyads (or a third of our anticipated sample of $n = 24$ caregivers). In the broader context, these data and the proposed model can be used to highlight important issues around financial burden for AYA cancer patients and their providers to develop targeted and flexible measures of financial burden for this unique cancer survivor group and to inform interventions that may alleviate the impact of financial burden for AYA cancer patients. Further, our future work will examine financial burden and its impact that likely extends into long-term survivorship for AYAs. We have focused on asking the questions that matter most to “give voice” to AYA cancer patients. The ultimate goal is for AYA patients to undergo cancer treatment and move into productive and full lives as post-treatment survivors in which their education and financial status may be less impacted from having had cancer at a young age.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

ACKNOWLEDGMENTS

The authors would like to thank the adolescents and young adult survivors and health care providers who generously devoted their time to teach us about their experiences with financial burden due to their cancer. We also want to acknowledge the research team from the Qualitative and Patient Reported Outcomes Shared Resource of the Wake Forest Baptist Comprehensive Cancer Center for their dedicated work on this project. Research reported in this publication was supported by the National Cancer Institute (NCI) of the National Institutes of Health under award number R01CA218398 (PI: Salsman). This research was also supported in part by the Qualitative and Patient-Reported Outcomes Shared Resource of the Wake Forest Baptist Comprehensive Cancer Center’s NCI Cancer Center Support Grant P30CA012197. Dr. Nightingale’s work on this manuscript was supported by the National Center for Advancing Translational Sciences (NCATS), of the NIH (UL1TR001420). The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Funding information

National Cancer Institute, Grant/Award Number: R01CA218398; Wake Forest Baptist Comprehensive Cancer Center, Grant/Award Number: P30CA012197; National Center for Advancing Translational Sciences, Grant/Award Number: UL1TR001420

DATA AVAILABILITY STATEMENT

Data are available on request from the authors.

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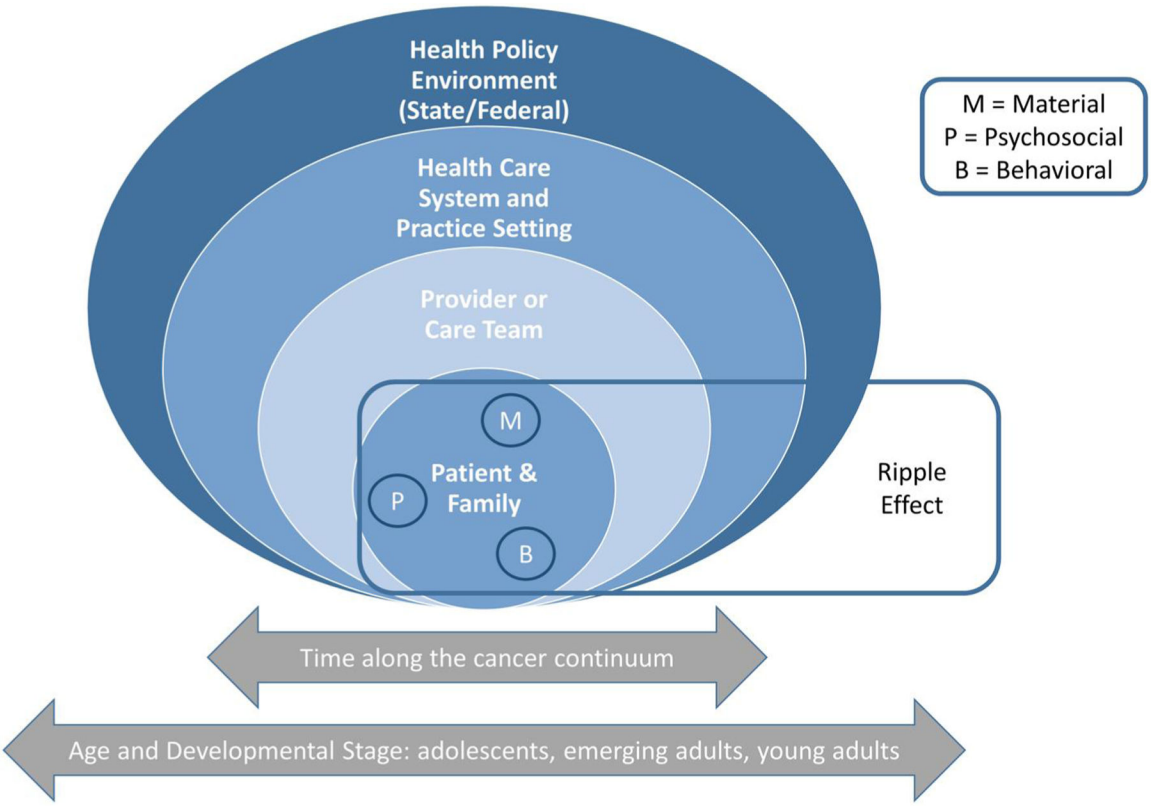

FIGURE 1.
Proposed conceptual model of financial burden

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TABLE 1

Stakeholder Characteristics[a] ($N = 72$)

	Adolescents (n = 10)	Emerging adults (n = 12)	Young adults (n = 14)	Providers (n = 36)
Gender				
Male	5 (50%)	8 (67%)	3 (21%)	9 (25%)
Female	5 (50%)	4 (33%)	11 (79%)	27 (75%)
Age (in years)				
Mean (SD)	16.5 (0.53)	21.17 (1.59)	33.07 (4.10)	45.70 (9.70)
Range	16–17	18–23	27–39	26–68
Race				
White or European American	6 (60%)	9 (75%)	10 (71%)	-
Black or African American	2 (10%)	2 (17%)	2 (14%)	-
Asian	0 (0%)	0 (0%)	1 (7%)	-
American Indian or Alaska Native	0 (0%)	0 (0%)	1 (7%)	-
Other	1 (10%)	1 (8%)	0 (0%)	-
More than one race	2 (20%)	0 (0%)	0 (0%)	
Ethnicity				
Hispanic/Latinx	1 (10%)	3 (25%)	4 (29%)	-
Cancer diagnosis (n)	Leukemia (3)	Leukemia (5)	Lymphoma (4)	-
	Lymphoma (2)	Lymphoma (1)	Breast (3)	
	Osteosarcoma (2)	Sarcoma (1)	Leukemia (2)	
	Sarcoma (2)	Adenocarcinoma (lung) (1)	Brain (1)	
	Germ cell (1)	Breast (1)	Cervical (1)	
		Melanoma (1)	Colorectal (1)	
		Testicular (1)	Testicular (1)	
		Thyroid (1)	Thyroid (1)	
Treatment status				
On	7 (70%)	6 (50%)	7 (50%)	-
Off	3 (30%)	6 (50%)	7 (50%)	-
Months since diagnosis				
Mean (SD)	7.90 (5.17)	27.58 (20.28)	17.07 (21.00)	-

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	Adolescents (*n* = 10)	Emerging adults (*n* = 12)	Young adults (*n* = 14)	Providers (*n* = 36)
Range	1–16	4–64	1–74	
Provider type (*n*, %)				
Physician	-	-	-	16 (44%)
Psychologist or counselor	-	-	-	7 (19%)
Social worker	-	-	-	6 (17%)
Nurse	-	-	-	5 (14%)
Navigator or coordinator	-	-	-	2 (6%)

[a]Number and (%) unless otherwise specified.

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TABLE 2

Summary of emergent higher-level themes for financial burden grouped by stakeholder category and conceptual model dimensions

	Stakeholder group				Dimensions of conceptual model		
Themes	**Adolescent**	**Emerging adult**	**Young adult**	**Provider**	**Material**	**Psychosocial**	**Behavioral**
Pre-cancer socioeconomic status and age at diagnosis impact financial burden on patients and families.	x	x	x	x	x		
Direct and indirect costs of cancer and its treatments are significant.		x	x	x	x		
Cancer creates a “ripple effect” of many indirect costs which further burden AYA patients.		x	x	x	x		x
Adolescents have limited awareness of the direct medical costs of cancer.	x				x		
Emotional impact of financial burden.	x	x	x	x		x	
Navigating the health care system is overwhelming.		x	x	x	x	x	
AYAs attempt to reduce financial burden by modifying cancer treatment-related behaviors.				x			x
AYAs adjust spending habits and stay organized to manage their finances.	x		x				x
AYA patients use a variety of psychological and behavioral strategies to cope with distress caused by financial burden.		x	x				x
AYA patients receive financial support and services from organizations.	x	x	x	x		x	x
The financial burden of cancer for AYA patients and/or their caregivers is long-lasting.				x	x	x	

Abbreviation: AYA, adolescent or young adult.